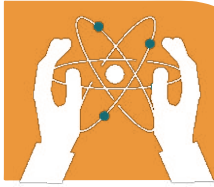


GLOSSARY

UNIT 01: CELL CYCLE : MITOSIS, AND MEIOSIS

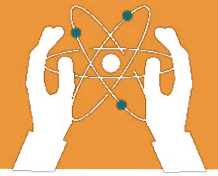
- 1. Cell:** The basic unit of life in all living organisms.
- 2. Nucleus:** The control centre of a cell that contains genetic material (DNA).
- 3. Cytoplasm:** The jelly-like substance inside a cell where organelles are suspended.
- 4. Chromosomes:** Thread-like structures made of DNA that carry genetic information.
- 5. Genetic:** Related to genes and heredity.
- 6. Mitosis:** The process of cell division that produces two identical daughter cells.
- 7. Meiosis:** The type of cell division that produces gametes (sperm and egg cells) with half the number of chromosomes.
- 8. Cytokinesis:** The final stage of cell division where the cytoplasm splits, forming two separate cells.
- 9. Significance:** The importance or meaning of something.
- 10. Phase:** A distinct stage in a process, such as the phases of cell division.
- 11. Repairing:** The process of fixing or restoring damaged cells or tissues.
- 12. Healing:** The biological process of recovering from injury or illness.
- 13. Reproduction:** The process by which organisms produce offspring.
- 14. Gametes:** Reproductive cells (sperm and egg) that carry genetic information for offspring.



FORCE AND ITS EFFECTS

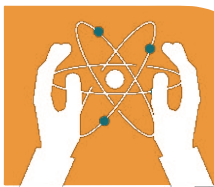
UNIT 02: HOMEOSTATIS:

- 1. Homeostasis:** The process of maintaining a stable internal environment in the body.
- 2. Feedback:** A biological mechanism that helps regulate processes in the body, such as temperature or hormone levels.
- 3. Internal:** Located inside the body or within an organism.
- 4. Regulation:** The control or maintenance of a process, often to keep balance in the body.
- 5. Transpiration:** The process of water loss from plants through small openings in leaves.
- 6. Hydration:** The process of absorbing or maintaining an adequate amount of water in the body.
- 7. Optimal:** The best or most favourable condition for function or performance.
- 8. Enzyme:** A protein that speeds up chemical reactions in the body.
- 9. Shivering:** An involuntary body response that generates heat by muscle contractions to maintain body temperature.
- 10. Dehydration:** A condition caused by excessive loss of water from the body.



UNIT 03: TRANSPORT IN LIVING ORGANISMS:

- 1. Transport:** The movement of substances, such as nutrients and gases, within an organism.
- 2. Photosynthesis:** The process by which plants convert sunlight into energy, producing oxygen and glucose.
- 3. Pathway:** A route or process through which substances move in the body.
- 4. Blood:** A fluid in the body that transports oxygen, nutrients, and waste materials.
- 5. RBCs (Red Blood Cells):** Cells in the blood that carry oxygen using haemoglobin.
- 6. WBCs (White Blood Cells):** Cells in the blood that help fight infections and protect the body.
- 7. Platelets:** Small blood cells that help in clotting to prevent excessive bleeding.
- 8. Plasma:** The liquid part of the blood that carries cells, nutrients, and waste products.
- 9. Veins:** Blood vessels that carry blood back to the heart, usually deoxygenated.
- 10. Arteries:** Blood vessels that carry oxygenated blood away from the heart to the body.
- 11. Deoxygenated:** Blood that has low oxygen content, usually returning to the heart.
- 12. Oxygenated:** Blood that is rich in oxygen, usually leaving the heart to supply the body.

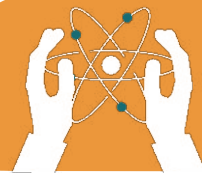


UNIT 04: FUNDAMENTALS TO CHEMISTRY:

- 1. Chemistry:** The branch of science that studies the composition, structure, properties, and reactions of matter.
- 2. Inorganic:** The study of compounds that do not contain carbon-hydrogen (C-H) bonds, such as metals and minerals.
- 3. Organic:** The study of carbon-containing compounds, primarily associated with living organisms.
- 4. Physical:** The branch of chemistry that deals with the physical properties and behavior of matter, including energy changes.
- 5. Biochemistry:** The study of chemical processes and substances that occur in living organisms.

UNIT 05: CHEMISTRY TERMINOLOGIES

- 1. Atom:** The smallest unit of matter that retains the properties of an element.
- 2. Molecule:** A group of two or more atoms bonded together.
- 3. Nucleus:** The central part of an atom that contains protons and neutrons.
- 4. Proton:** A positively charged subatomic particle found in the nucleus of an atom.
- 5. Neutron:** A neutral (uncharged) subatomic particle found in the nucleus of an atom.
- 6. Electron:** A negatively charged subatomic particle that orbits the nucleus of an atom.
- 7. Elements:** Pure substances consisting of only one type of atom.
- 8. Compound:** A substance formed when two or more different elements chemically bond together.
- 9. Heterogeneous:** A mixture in which the components are not evenly distributed and can be distinguished.



10. Homogeneous: A mixture in which the components are evenly distributed and appear uniform throughout.

11. Chemical Equation: A symbolic representation of a chemical reaction using chemical formulas.

UNIT 05: CHEMICAL REACTIVITY:

1. Metals: Elements that are typically shiny, good conductors of heat and electricity, and malleable.

2. Non-Metals: Elements that are usually dull, poor conductors of heat and electricity, and brittle.

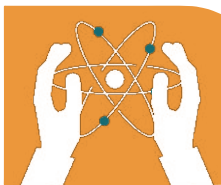
3. Metalloid: Elements that have properties of both metals and non-metals.

4. Conductor: A material that allows electricity or heat to pass through it easily.

5. Alkaline Metals: Highly reactive metals found in Group 1 of the periodic table, such as sodium and potassium.

6. Halogens: Highly reactive non-metals found in Group 17 of the periodic table, such as fluorine and chlorine.

7. Reactivity: The tendency of a substance to undergo a chemical reaction, either by itself or with other substances.



12. Reactant: The starting substances in a chemical reaction that undergo change.

13. Product: The substances formed as a result of a chemical reaction.

UNIT 07: INTRODUCTION TO KINEMATICS:

1. Kinematics: The branch of physics that studies the motion of objects without considering the forces causing it.

2. Motion: The change in position of an object over time.

3. Force: A push or pull acting on an object that can cause a change in motion.

4. Translatory: Motion in which all parts of an object move in the same direction and distance.

5. Rotatory: Motion in which an object moves in a circular path around a fixed axis.

6. Vibratory: Motion in which an object moves back and forth around a fixed point.

7. Scalar Quantities: Physical quantities that have only magnitude (size) but no direction, such as mass and speed.

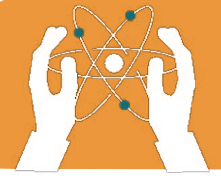
8. Vector Quantities: Physical quantities that have both magnitude and direction, such as velocity and force.

9. Distance: The total length of the path travelled by an object, regardless of direction.

10. Speed: The rate at which an object covers distance, measured as distance divided by time.

11. Average Velocity: The total displacement divided by the total time taken.

12. Displacement: The shortest distance between an object's initial and final position, with direction.



UNIT 08: INTRODUCTION TO DYNAMICS:

1. Dynamics: The branch of physics that studies the forces and their effects on motion.

2. Force: A push or pull acting on an object that can cause it to accelerate or change direction.

3. Mass: The amount of matter in an object, usually measured in kilograms (kg).

4. Velocity: The speed of an object in a given direction.

5. Acceleration: The rate at which an object's velocity changes over time.

6. Deceleration: Negative acceleration causing a decrease in velocity over time.

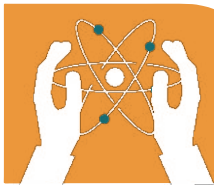
7. Friction: A force that opposes the motion of an object when it is in contact with a surface.

8. Newton: The SI unit of force, named after Sir Isaac Newton.

$$(1 \text{ Newton} = 1 \text{ kg} \cdot \text{m/s}^2)$$

9. Weight: The force exerted by gravity on an object, calculated as: **mass × gravitational acceleration ($W = mg$)**.

10. Inertia: The tendency of an object to resist changes in its state of motion unless acted upon by an external force.



UNIT 09: FORCES AND THEIR EFFECTS:

1. Hydraulic Lifts: Machines that use pressurized fluid to lift heavy objects by applying Pascal's principle.

2. Stretched: The action of elongating or extending an object due to an applied force.

3. Deformation: The change in shape or size of an object due to the application of a force.

4. Stiffness: The resistance of an object to deformation when a force is applied.

5. Pressure: The force applied per unit area ($P = F/A$).

6. Area: The surface over which a force is applied or distributed.

7. Fluid: A substance that can flow, such as a liquid or gas, and has no fixed shape.

8. Density: The mass of an object per unit volume ($\rho = m/V$).

9. Depth: The distance from the surface to a lower point, often affecting pressure in fluids.